CONFIGURING LIVENESS AND READINESS PROBES IN KUBERNETES

Kubernetes relies on the health probes (liveness and readiness) to accomplish its tasks as a container orchestrator. For example, when the desired state of an application is to have three replicas, Kubernetes ensures there are always three application instances running. If any of them doesn't return a `200` response from the liveness probe, Kubernetes will restart it. When starting or upgrading an application instance, we'd like the process to happen without downtime for the user. Therefore, Kubernetes will not enable an instance in the load balancer until it's ready to accept new requests (when Kubernetes gets a `200` response from the readiness probe).

Since liveness and readiness information is application-specific, Kubernetes needs the application itself to declare how to retrieve that information. Relying on Actuator, Spring Boot applications provide liveness and readiness probes as HTTP endpoints. Let's see how we can configure Kubernetes to use those endpoints for the health probes.

In your Catalog Service project (catalog-service), open the Deployment manifest (k8s/deployment.yml), and update it with configuration for liveness and readiness probes as follows.

Listing 13.11 Configuring liveness and readiness probes for Catalog Service

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: catalog-service
  ...
spec:
  ...
  template:
    ...
```